MIXERMAN'S

ULTIMATE

GUIDE

TO

PRODUCING

RECORDS, MUSIC & SONGS

THE ANTIDOTE TO MISINFORMATION



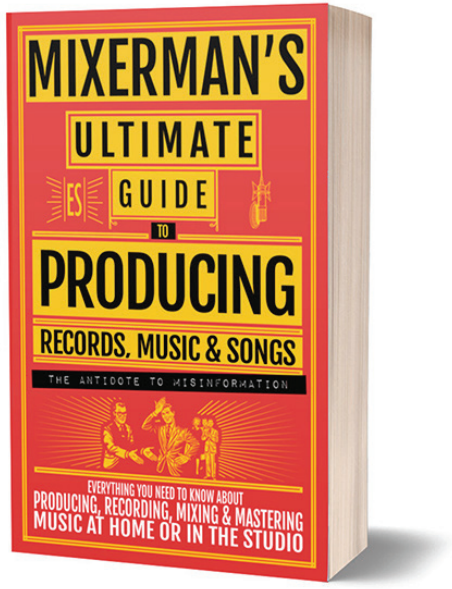
EVERYTHING YOU NEED TO KNOW ABOUT

PRODUCING, RECORDING, MIXING & MASTERING

MUSIC AT HOME OR IN THE STUDIO

MIXERMAN'S ULTIMATE SAMPLER

Selections from:



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Introductions

Excerpt

to learn everything that I share with you in this guide. It takes years of dedicated focus across a varied set of circumstances to master any one discipline. It's absolutely insane to try to master them all at once. But this is now the world that we live in, and the reason for this book.

My Producing Journey

I started piano at the age of 6. I was classically trained, and by the time I was a senior in high school, I was performing Chopin and Rachmaninoff works—which are some of the most challenging piano works to play. I picked up the trumpet at 8, was first chair in my High School Wind Ensemble and lead trumpet for the jazz band.

In 1985 I went to Rutgers University to study jazz piano with Kenny Barron, one of the world's great straight-ahead jazz pianists. I was relatively new to jazz, and it didn't take long to realize I wanted to work in more contemporary music. I also didn't see myself as a professional player. I definitely wanted to be in music. I just didn't really know how yet.

The next year, I transferred to Berklee College of Music in Boston, where I studied Songwriting among other things. I really had no intention of graduating—I didn't need a degree to make my mark in music, had no interest in teaching, and felt back-up plans were prone to back*firing*. So, I changed my major four times just so that I could take specific classes that interested me most.

After my first year at Berklee, I landed at Dimension Sound, a 24-track analog studio in Jamaica Plain, a suburb South of Boston. This was where George Thoroughgood had recorded several of his albums, including his iconic hit "Bad to the Bone."

I lived above the studio and had 24-hour access to the room. I knew almost instantly that was where I belonged. The studio was home to me. I also had mentors to teach me and spent the

next four years cutting my teeth as a record maker and engineer, both on my own songs and for outside clients. It took me that long before I was even "good" at any of it.

By the time 1991 rolled around, I made the move to Los Angeles. I wanted to work on big records, and the best way to do that was to live in close proximity to the industry. And although I grew up an hour outside of NYC, I couldn't take the cold any longer. Los Angeles it was.

My first gig upon arriving in Los Angeles was on the Setup staff at Capitol Studios, where I learned how to set up large orchestral sessions. This was largely a glorified runner gig, but there was plenty of setting up and tearing down to do. The biggest acts of the time came through Capitol, and I learned a lot during my six-month stint there.

At the end of 1991, I left Capitol and took a gig as a house engineer at Hollywood Sound. The engineers' room was full of rock guys, and none of them were particularly interested in working on hip-hop. So, I took most of the available hip-hop gigs. Not only did I enjoy hip-hop, but I also much preferred working as a recordist rather than as an assistant.

A month or two in, I was assisting on a session for the Pharcyde, who were having their first single mixed in Studio B. I was in the back of the room with Mike Ross, the President of Delicious Vinyl, as he vented about how much he disliked recording at a studio down the road. I asked why he didn't just record at Hollywood Sound. Apparently, he didn't want to spend $800 a day just to record vocals. This I could understand. I also had a solution.

The owner of Hollywood Sound, Jesse Hodges, had installed a production room upstairs, complete with a Spectral Sonics DAW. Mind you, this was long before DAWs had even begun creeping into studios—very few producers were using them. The industry hadn't even gone through its ADAT phase yet. Jesse wanted $35 an hour for the room and the DAW, the same rate

Mike was paying at the other studio. So, I showed him the room, introduced him to the DAW, and convinced him to record the album there—with me, given my experience as a recordist.

The next evening, I was recording vocals with the Pharcyde. They really only knew me as an assistant, and one of them got the bright idea to hold the Neumann U47—which, for those who don't know, is not designed to be held. It's a large, sensitive mic, usually suspended in a shock mount. Naturally, I told them there would be an inordinate amount of rustling noise, booms and pops, making their vocal recordings largely unusable. I even demonstrated it on playback. Somehow, they didn't seem to mind it. Regardless, I wasn't going to tell them "No." This was a test of my loyalties.

Mike approached me the next evening, as if to relieve me of my duties. Clearly, he'd heard the roughs from the previous night's recordings. Sensing urgency, I quickly explained what had happened.

"*They held the 47!*"

He was incredulous. "Why would you let them do that!"

"Because if I had stopped them on our very first session together, they wouldn't trust me," I explained. That seemed to satisfy him. He grabbed a scrap of paper, jotted down his car phone number, and then called the group into the room for a meeting.

"No more holding the 47!" he bellowed. Then, as he turned to leave, he overtly handed me his number, looked at the group, and yelled again, "And you need to listen to Eric! He knows what he's doing!"

I spent the next eight months recording *Bizarre Ride II the Pharcyde*, which became my first gold record—and ultimately, one of the quintessential hip-hop albums of all time. Proximity clearly had its benefits, as did practice and preparation. I was paid $120 a day to record that album, which wasn't much even

back then. Whatever. Success begets success in this industry, and that album paved the way toward my future.

I parlayed the success of that Pharcyde album into several years of mixing hip-hop, soul, and acid jazz albums—almost exclusively. I went fully independent in 1993 while mixing The Brand New Heavies *Brother Sister* album. In fact, Vicki Giordano, the traffic manager at Hollywood Sound, let me go as staff right in the middle of mixing it. Perhaps as a favor—because on Wednesday, I was making $10 an hour as a staff engineer. On Thursday, I was making $35 an hour, paid directly by the label, for the exact same work.

I loved mixing these genres. They were fresh, and I enjoyed the music. Of course, I didn't have much choice in the matter—those were the gigs that came my way, so those were the gigs I took. Lots of them.

Even so, I had concerns about being pigeonholed as a hip-hop mixer. I wanted to work across a diverse range of musical styles, and becoming too entrenched in one genre had its pitfalls. My ultimate goal was to be a producer, and as much as I liked hip-hop, I was a cultural outsider—I wasn't a beat-maker, nor did I want to be. That put a cap on how far I could advance within the genre, and to me, that was a problem.

Then, as if on cue, opportunity knocked.

In the summer of 1995, I got a call from Andy Factor at Virgin Records to mix Ben Harper. Michael Andrews—the guitarist for The Greyboy Allstars, whom I'd worked with on a Karl Denson album collaboration—had personally recommended me for the gig. Michael is now a somewhat notable film composer.

Andy was calling me about Ben's upcoming sophomore release, *Fight for Your Mind*. Apparently, they had already gone through three mixers, all of them stunningly big names in rock music—all of them summarily dismissed. I literally laughed out loud when he told me the name of the last mixer they'd fired—that's how shocking it was to me.

Introductions

Excerpt #2

the sound. Every decision—your arrangement, your recording choices, the quality of the performances you capture, how you think about space, instrumentation, and musical function, how (and why) you use processors and plugins, how you apply motion and shape distortion, even your approach to artistic refinement—all of that falls under sound design. The sound is so deeply intertwined with the song and the music, that by the time you reach the *Mixing & Mastering* chapter, you'll already have everything you need to set you up for a great mix.

I structured this book in a way that allows producers at nearly any level to follow along, other than the absolute beginner. Even so, you might occasionally come across a term before I've had the chance to fully flesh it out. This was unavoidable. Had I stopped to define every technical term the moment it appeared, this book would be unreadable. So, if you run into an unfamiliar term, *look it up*—factual information in regard to technical terminology is just a quick search away.

Of course, the factual technical information that I offer throughout isn't what sets this book apart. Anyone who has experience engineering music for enough time can relay that to you. My early books barely touched on it. The musical and practical decisions that shape a record are of far more value than understanding the technical ones, and that's what my books have always addressed first and foremost.

In my first version of *Zen and the Art of Mixing*, I don't even talk about how to control a compressor. Which may seem baffling, but learning how to use a compressor really just requires some practice. In the early part of my career, I was mixing records on a daily basis, including gold ones, yet I couldn't have described to you specifically how to control a compressor. I just twisted knobs aggressively until I heard something I liked. And if that didn't work? I'd try a different compressor.

In fact, I wore my lack of technical expertise as a badge of honor. My processing tools were not technical instruments to

me, but rather musical ones meant to be pushed well beyond their limits. I wanted to break them in the process. I couldn't care less what those timid engineer-types considered optimal. Frankly, I took offense to being called an engineer. *Mixer!* I would retort.

As far as I was concerned, I was an artist, just like the artists and producers who hired me to shape their sound and maximize their impact. The gear was merely there for me to mangle and control the sound. I made my decisions purely by ear, often relying on hit-or-miss strategies.

Over time, of course, I figured out what all the knobs did and how the controls worked. When I transitioned fully to digital—which is far less forgiving than analog—I was almost forced to learn the technicalities. Writing about it forced that too. And while that knowledge gave me more control and at times made the process more efficient, it certainly didn't make me a better mixer or producer.

Worry not—I can, and do, explain how to control our most important processors, breaking down the technical side in a way that's easy to understand and digest. It's the technical considerations that trip people up, especially in a world where mentoring has gone to die, and bad information is spread around online like the plague. That's why this guide is so comprehensive—because you need reliable, accurate technical knowledge. But make no mistake—the real value lies in the perspective I offer you.

This entire book comes from my unique perspective as a successful professional music-maker with a diverse discography—one that spans decades of changes in technology and musical trends. I don't just explain *how* to produce, but *why* certain decisions matter. And not just why *I* make certain decisions, but why *you* might make certain decisions—whether musical, technical, organizational, monetary, or even political in nature. All of it based on real-world practicalities. That's the good stuff.

There is information and insight in this guide that you just won't find anywhere else.

I will often offer you my preferences to go along with my explanations, all of them based on strategy within the context of musical decisions. There is no strategy that works for all people in all situations. There is only the strategy that best fits the moment, the music, the talent, your goals, and your preferences. Over time you'll develop your own strategies, based on your own experiences and needs. You may even reject my advice at times, which you should. That's one of the more effective ways for you to eventually realize I'm right.

Ahem.

There is no technique in producing music that could be considered invalid. If it works, it works. As such, you'll get my opinions, but you'll find very little dogma on my part. I seek to arm you with strategies to help *you* achieve *your* vision, so that you can create music and songs with intent, exactly as you hear it.

Ultimately, music is meant to evoke feelings from great songs. There is no right or wrong where it comes to the creation of art. There is only your intent, how easily you express that intent, and how you use it to connect with an audience. Nothing else matters.

That's what this book offers you. Whether you're a semi-beginner producing your own songs or even a somewhat seasoned Producer For-Hire. I've discussed everything I could think of in this guide—all of it designed to help you to work in a more effective and creative manner. This is true whether you do everything yourself at home or manage a team in the studio.

Given the broad range of the information, you may come across sections that you feel don't pertain to you. Far be it for me to tell you how to read a book, but I would advise against skipping sections, even if you feel they have little relevance to your circumstance. Even as a self-producer, understanding

every role in the process will make you better at producing your own work.

Surely you can jump around and try to derive the information that you feel applies directly to you, but you will surely miss out on countless insights that I weave in throughout the book. Like I said, this is not a linear process, and concepts that can assist you may come from unexpected places.

If you want to be the most effective producer you can be, you need to know everything. That's the point of the book after all. To explain *everything*.

Now that we've gotten the introductions out of the way, let's dive in.

The Producer

Chapter One Excerpt

The Songwriter Producer

When I wrote *Zen & the Art of Producing* back in 2012, the songwriter was one of the weakest positions from which to become a producer. Now it's by far the strongest.

In the olden days, when records were still primarily made in dedicated commercial studios, Songwriter Producers were generally relegated to demos, which they delivered to a publisher to shop. There was a psychology to a good demo, and it usually entailed dumbing-down the product to attract both an artist and a producer. The worst thing a writer could do is turn in a demo so fantastic that no producer dared to touch the track for fear of failing to improve upon it. As such, songwriters often developed all the wrong skills.

The modern DAW has changed all of that. Between perfect recall, and the ability to program and record tracks at home, the songwriter had more time to develop their production skills. The two jobs have become nearly synonymous in the pop production world. If you write in any of the pop music genres, you probably also produce your records, and this is true at the earliest stages of your development.

Songwriter Producers are often directly involved with every facet of the record from inception to completion. They write the song, program the track, record the singer (the artist), record parts, and may even mix the record, regardless of whether they're currently qualified for such tasks. Of course, you become qualified at something by repeatedly doing it.

For many Songwriter Producers, the song and the production are of equal importance. You can't present the song in a compelling manner without the production. They are inexorably attached. Without a production, a song is merely an idea, and to bring that idea to fruition, there must be a production.

Most Songwriter Producers today write and program pop music, which is a genre that all but requires a somewhat fresh

production. Please understand, I use the term "pop music" here in a very broad manner. You can classify a genre as anything you like—R&B, electronica, house, hip-hop, reggaeton—if the production is mostly programmed and meant to appeal to a large array of listeners, that's pop music where it comes to production.

Oddly, the best way to achieve a fresh production is to use trendy production techniques, which admittedly may seem somewhat counterintuitive, but consider the alternatives. You can use techniques that are wholly out of fashion which will result in a stale production from an undefinable era. On the other end of the spectrum, you can create something so unique and outside of the mainstream that nobody will even understand it. The goal in producing a popular song isn't to make something wholly new. The goal is to make something that's mostly familiar but comes off fresh.

To be clear, "retro" productions that are meant to remind us of a rather distant era of music are always in fashion to some degree, and can still sound modern, to boot.

Of course, you can make a production sound downright silly if you employ every trendy production technique there is, regardless of whether it fits the song or not. Which is why the Songwriter Producer is such an effective position from which to operate. If you write a somewhat trendy song, then a modern production will often fit perfectly. And since the song and the production tend to be developed simultaneously, the Songwriter Producer can more easily push trends into new territory.

Songwriter Producers, particularly those who have had successful songwriting careers, are exceptionally strong when it comes to working with solo artists lacking material and can be desirable for bands or artists in need of help improving their songs.

It only takes but one hit song to make your entire career. And while nobody wants just one hit, the payoff of a popular

record can be so great for the songwriter, that it makes sense to invest time into your craft before a budget ever even exists. As a songwriter, it's critical to develop your arrangement and sound design chops for purposes of producing your songs. Either that, or you need a partner who can cover that side of things.

There is no budget for writing your song, because as a songwriter, you need to write, program and record as many songs as possible. This is how you attract an artist to your song. You must invest time in your "demos" because these will ultimately become the record. As such, the Songwriter Producer is often working in a speculative manner.

If there's any stigma to be had for the modern-day Songwriter Producer, it's that their skill set is often limited to programmed material. That can make you rather one-dimensional, which is always fraught in the music business, because fads, trends, and production styles are guaranteed to change. Meanwhile, you're perceived as outdated.

The Musician Producer

The Musician Producer comes from a musical background. Some musicians migrate to producing because their run of success has either ended or peaked. Some transition to producing because it interests them, or because they're tired of touring. Regardless of the reason, those known as Musician Producers are typically highly proficient at an instrument (maybe several) and are often hired because of this. There is much respect in the music industry for well-regarded musicians and artists who become producers, making this one of the better positions from which to transition.

Musician Producers aren't viewed as having engineering skills, although that's not really an issue since most producers need to perform many of the engineering tasks these days. Musician Producers who are hired largely for their prowess on an

Songwriting & Arranging

Chapter Two Excerpt

beat could even serve as both rhythm and countermelody. A tuned 808 KIK drum can serve a rhythmic and bass function.

As you produce your arrangement, it's important to understand what functions any given part is serving, and to evaluate what's missing from your production that might help you with forward motion. If your prechorus isn't hurling the listener into the chorus as it should, then one possible solution is to bring in a countermelody to create tension for purposes of release at the chorus. Or you could add a rhythmic part for purposes of acceleration.

Mixerman's Five Planes of Space

It may seem strange that I talk about space in terms of your arrangement, but that's precisely what we're dealing with here. Think about it. We can absolutely consume a production with low-end frequency information. That would be space. Our panning is limited to the width of our monitors. Also space. We balance our parts in the production, and whatever is placed loudest will appear closest and most up front. Space. Reflectivity, which is achieved through the addition of spatial information (reverbs and delays), creates the illusion of distance and depth. Space. And finally, we use the contrast between sparse and dense sections to create a contrast dynamic that reveals itself over time. Space!

Of course, where it comes to space and time, we do have limitations because it's contained. We work within a defined range of audible frequencies. The width of our production is limited to the extent of our sides. Our balances require some measure of proportionality in order to maintain the illusion of depth. And you can only cram so many things into a dense production before you lose all contrast and find yourself unable to decipher all of the information.

Given that we are operating on what amounts to a sonic canvas, we must be mindful of our space and how we use it. And so, I have defined the following as my Five Planes of Space.

In a stereo image there are five planes of space that we use to create a four-dimensional image (the fourth dimension being time): panning—left to right; frequency—up to down; balance—front to back; reflectivity—far to near; and contrast (dynamics)—sparse to dense, bright to dark, asymmetrical to symmetrical.

Panning—Left to Right

The strongest stereo image is derived from two wholly independent mono sources panned hard left and hard right. In the case of guitar (for instance), the strongest image will come from two unique players on their own instruments performing their own unique parts. This is a stronger image than one produced by one player performing a true double with the same guitar.

Why is that?

Musicians who have reached at least a modicum of proficiency on their instruments tend to play with some consistency, which creates a subtle sonic fingerprint. If I have the same player on the same guitar perform a true double, panned hard, there will be enough timing differentials to produce a strong stereo image. There will also be subtle timing and tonal similarities. As such, the guitars will not appear quite as wide as they might were we to switch some things up. Changing out the guitar on the double will serve to slightly widen the image.

The weakest stereo image (aside from most keyboard patches) occurs when you employ stereo miking techniques, mostly because you're collecting both side and center information. It's still a stereo image, but the two sides are not wholly independent, as such, they form an image that spreads across the entirety of the field, including the middle.

I routinely pan stereo organic drum overheads hard left and hard right because the stereo image isn't all that wide to begin with. Meanwhile, others on the Internet proudly proclaim that they toe their overheads in, meaning they soft pan them. But all that does is cram even more sonic information in the middle, which both reduces the apparent size of the kit, and increases the competition for space in the center of your production.

When you pan two independent parts such that one sounds only through the left monitor and the other only through the right, you have defined the edges of your width plane. Make no mistake, you will make your life easier if you use the full width of your stereo field. Why? Because when you muck up the middle you eat valuable space.

When you toe in your two guitars to internal pan positions rather than to pan them hard, and when you bring in your drum overheads rather than to pan them hard, you are purposely choosing to reduce the space within which you have to operate. This would be like moving all of your living room furniture off the wall by two feet. You've accomplished nothing other than to reduce your usable space.

From the perspective of the listener, it makes no difference where you pan anything. None at all. For starters, Music Fan really doesn't care. You will never hear a punter complain about where things are panned within a production. They might muse about it on rare occasion (usually positively), but your panning decisions won't ever cause someone to shut off your record.

The fact of the matter is, unless the listener is wearing headphones, they are usually located outside of the stereo image. We can't even really consider headphones in terms of imaging, because the closed ear nature of them prevents the two sides from interacting, which is a skewed and unnatural image that doesn't align with the three dimensional manner in which we actually hear. When you play music through monitors the music interacts within a space. Not so with headphones.

Songwriting & Arranging

Chapter Two Excerpt #2

Really, the way that your parts fill the frequency spectrum in an arrangement is equally important to their musical function. Frequencies are the building blocks of music, and as such, we'll spend quite a bit of time on the subject. In the meantime, there are more planes of space to discuss.

Contrast—Perceived Dynamics

Whereas we can listen to a mix and immediately hear frequency, panning, and balance decisions, we discover contrast over the course of the record—that is to say–time.

Contrast is used to create the illusion of dynamics in modern music. Dynamics, by definition, is: a variation in force or intensity, especially in musical sound. Unfortunately, dynamics in music have been greatly reduced over the years. Generally speaking, we've completely eradicated tempo fluctuations from modern music, (which affects intensity), and many records use very little of the available dynamic range from the loudest to the quietest parts of a mix (which we call loudness).

That said, we do often listen to music in noisy environments, and too much dynamic range can be a problem if you ever hope to hear your record well in the car, or while the dishwasher runs, or the vacuum, or any external noise within your vicinity. This means we need to create the illusion of a dynamic in other ways, and we accomplish that with contrast.

When I mixed Ben Harper's "Roses From My Friends" (from his third album, *Will to Live*), I originally sculpted an enormous dynamic range for that mix. The chorus was quite exciting given how much louder it was than the verses. Unfortunately, once I got the mix into my car, I found myself tuning out until the chorus kicked in. And if I started the record at a level such that I couldn't ignore the verse, then the chorus tore my head off. What was an exceptionally effective payoff in the sound-isolated confines of the studio, was too dynamic for a proper payoff in

the real-world environment of a car. Given this, we brought the verses up in level in relation to the choruses.

Now, while we had to reduce the overall dynamic range of that mix in order for it to work in the outside world, we still had contrast in our favor. The verse is sparse and sweet, while the chorus is dense and aggressive, so we get the illusion of more dynamic range than actually exists. Not only do dynamics offer contrast, but contrast offers us the perception of a dynamic in its own right, which can increase payoff.

Then there's symmetrical contrast. We already touched upon this, but a single guitar in the verse panned hard will contrast nicely with double guitars panned hard in the chorus. If we add a Rhodes in its lower register and place it in the verse opposite the guitar, we will now have stereo symmetry in the verse, which will greatly minimize the symmetrical contrast we had earlier. Where we've lost, we've also gained. There is now a contrast in frequency that occurs between the dark Rhodes in the verse and the presumably bright, gritty electric guitars that take over in the chorus. It doesn't matter from where you derive your contrast, so long as it exists.

That said, contrast, while a definitive plane of space, is nothing more than a tool. Just as you can mix a song in mono, and thereby reject all that great stereo width, you can choose to use little to no contrast within your arrangement. Songs can have their own natural contrast, which you can choose to mirror, or allow to stand on its own.

Reflectivity—Far to Near

Reflectivity is the illusion of space. Whereas we accomplish the three-dimensional illusion of width, height, and depth through panning, frequency, and balance respectively, we accomplish the illusion of reflectivity within all three dimensions plus the fourth—time.

The Home Producer

Chapter Three Excerpt

Sadly, no.

Here I was—an expert at mixing, recording, and producing—yet my skill set was all but eradicated by the acoustic realities of the room I was working in. Turns out, there's a reason those expensive studios invested so heavily in acousticians, designers, and proper buildouts.

Acoustics

The first step to fixing a problem is acceptance. As long as you convince yourself that you can simply compensate for a problem room, you will struggle.

Let me preface this section with this: My purpose isn't to dissuade you from working with what you have or to suggest that you have no shot at becoming a professional just because you're learning your craft in a compromised space. You *can* make great music regardless of your acoustic environment. Great music has been made on the streets with nothing more than a mic and a boombox. The classic Motown records from the '60s were recorded in a house with almost primitive acoustic treatment. Don't think for a moment that compromised acoustics will prevent you from achieving musical greatness—or even success, for that matter.

If you write and produce an amazing song, and a major-label artist wants to sing it, you'll be placed in a commercial studio with highly skilled professionals to bring your vision to life. Your musical skills matter far more than the engineering. Successful songwriters can make millions from just *one* song. Successful engineers get paid by the hour. So, you tell me which has more value.

That said, I can certainly understand your quest for a professional sounding product and the desire to work in monitors rather than headphones. If you find yourself constantly frustrated because your sonic vision never seems to match the

results, and if you struggle to capture your source the way you hear it in your head, or if your mixes never translate as expected outside your space, then the only way to fix those issues is to deal with the acoustics.

If you know exactly how you want your record to sound, only for it to sound completely different outside your room, that's an acoustics issue. If your balances, EQ curve, and spatial presentation are all out of whack, then the sonic decisions you're making in your room are flawed. What you're hearing from your monitors is a distortion of reality. It's a lie.

You can't just reference your way past a lie—because the reference is a lie too. And sometimes, it's a lie of omission, like when an entire low-end frequency band disappears due to standing waves.

If you don't fix your room—and every room that isn't purpose-built for mixing and recording needs some level of fixing—you'll have a monumentally difficult time delivering tracks the way you sonically envision them.

The Challenge

I suppose this would be about the point that I should reveal to you exactly how to make your room perfect. Unfortunately, that entirely depends on the specific realities of your space. Every room presents its own challenges, and some spaces are nearly impossible to make acoustically accurate for proper translation in the outside world.

Small rooms with relatively low ceilings are an absolute acoustical nightmare. Low ceilings, in general, create severe acoustic problems, and even 8-foot (about 2.4-meter) ceilings—which are fairly common—are considered low when it comes to both monitoring and recording.

As long as those issues exist, you'll have to take your production outside of your room to find its flaws, and that's an exceptionally inefficient way to work.

I've seen people ask for help treating all kinds of problematic rooms—L-shaped spaces, long narrow rooms that resemble hallways, square rooms, bedrooms where they sleep, rooms with closet cutouts, ceilings under seven feet high, even spaces with chandeliers or ceiling fans creating unpredictable reflections. Of course, ceiling fans can be easily removed, but solutions for some of those other problems can be challenging at best.

These frustrated producers genuinely believe they can fix fatal acoustical flaws—often for little-to-no money—or that they can simply learn their space over time and compensate for its deficiencies. Nothing could be further from the truth.

Of course, I'm telling you all this as someone who is not an acoustician. I have no formal credentials in that regard. Everything I know comes from experience, backed by practical knowledge. It's the result of a long and difficult transition from highly regarded, outrageously expensive mix rooms, to a converted room in my house—several rooms, actually.

I've had to muddle through these issues just like everyone else. I too firmly believed I could make any old space work, that I could reference my way to an understanding, and that my skills would overcome even the most fatally flawed acoustics.

Then I tried it.

Fail. Not once. Multiple times.

What's particularly baffling—as you dig deeper into the rabbit hole of acoustic treatment—is the wildly conflicting opinions between acousticians. Which is strange, considering this stuff is supposed to be based on math. It really makes me wonder how anyone is supposed to figure this shit out when even self-proclaimed experts can't seem to agree. Of course, here I am—a total hack when it comes to acoustics (at least I know it)—giving you advice on how to fix your situation in the abstract.

What could possibly go wrong?

And while I've spent countless hours learning what I can about acoustic practicalities, offering advice on the subject is nothing short of fraught. My failures have been limited to my own rooms, which gives me an inadequate knowledge base to claim any expertise. So I won't stray too far out of my lane in this regard.

What I *can* offer is concrete information on curating a more accurate monitoring environment. Some of it comes from extensive reading on studio acoustics, and some of it comes from firsthand experience correcting my own rooms over time. Take what I provide here and supplement it with guidance from reputable sources.

Positioning

First, if your room is rectangular, it's recommended to position your monitors along the short wall. This allows for the greatest possible distance behind your listening position, which helps with more accurate low-end response. Even in relatively narrow rooms, this orientation is usually the best approach.

It's commonly recommended that you place yourself at 38% of the room's length and your monitors so they form an equilateral triangle with your listening position. If you end up sitting at 50% of the room's length, you're right in the middle of a major problem—standing waves. This can cause certain low-end frequencies to cancel out while boosting others, making it impossible to get an accurate read on your mix.

Don't get mad at my use of the word *impossible* here. It's not pessimism—it's just reality. It's also impossible for a human to jump 100 feet (about 30 meters) in the air on Earth, yet nobody would take issue with that statement. Optimism and encouragement won't change the laws of physics. They'll only cause you to believe something that isn't true.

The Home Producer

Chapter Three Excerpt #2

ear cups. If you plan to work extensively in headphones, both comfort and accuracy are worthwhile considerations.

That said, not all of your work requires perfectly accurate monitoring. No matter how inaccurate your room, it won't prevent you from writing a great song or arranging a compelling production. It won't stop you from compiling a vocal from multiple takes. It only limits your ability to make a mix that translates beyond your environment. If you're in a space that can't realistically be made accurate, apply enough acoustic treatment for an accurate recording capture, and rely on high-quality headphones for accurate mixing.

At the end of the day, great music is about performance and arrangement, not just the mix. You can produce an incredible song in headphones, even without an ideal monitoring setup. But when finalizing a mix, a second reference—whether a treated room, trusted monitors, or even checking on consumer playback systems—can assist in achieving your goals.

If headphones are your primary mixing environment, investing in a quality reference pair and using crossfeed or virtual monitoring tools will go a long way toward making your mixes translate better.

Distortion

Before we discuss our various recording and mixing processing tools, I'd like to take a moment to address distortion. Throughout this book, I'll frequently reference distortion—its properties and how it factors into sound design—so I want to make sure I don't give you the wrong impression or promote misunderstandings as we progress.

First, distortion does not necessarily mean aggressive, overt, or obviously degraded sound. At its core, distortion simply means to alter or change something, ostensibly in an "unflattering" way. But in music and art, unflattering is often what

makes something compelling. Picasso's paintings wouldn't be valuable if they weren't distorted representations of reality. We love distortion because it adds character, which makes things more interesting, textured, and emotionally engaging.

Really, the question isn't whether you *should* distort. Distortion is a critical component to record production. The question is really a matter of *how much* distortion and on what, and that's not an arbitrary decision. You can distort overtly such that everyone would recognize it as distortion, and you can distort subtly such that you would need to compare to the original in order to recognize the distortion. But make no mistake, distortion is your friend.

As I discuss strategies for setting levels on analog mic preamps and compressors, you might begin to wonder whether I distort everything all the time. Which I don't. Certainly not in an audible manner. But it's an important consideration, and it's beneficial to understand how to use distortion to your advantage.

When I worked in the all analog domain, low distortion specs were touted, because you wanted to keep your distortion in control. These days, many Home Producers are using virtual interface preamps, which often lack in useful distortion properties. This makes externally derived distortion an important part of your sound design. If you're not deriving your distortion at the time of recording, then you need to shape your distortion profile using plugins. Whereas we once worked to minimize distortion, now we have to be deliberate in how we apply it—because a production without the right kind of distortion can sound sterile and lifeless.

Distortion Purposes

Distortion does not always present itself as breakup. Distortion can merely be a slight thickening of the tone that occurs

from an analog preamp or compressor. Aggressive overcompression is essentially a type of distortion. How else would you describe heavy pumping effects, or the sound of harmonics that resonate as you put a part into a steady state?

There are specific reasons to strategically introduce distortion to your parts and production. Those being: agitation, edge, thickening, resonance, sustain, obfuscation, or clarity.

Agitation

If you want to agitate the listener, overt, aggressive distortion is a great way to do it. This kind of distortion is a staple of industrial and punk music, especially on '80s punk records. Personally, I don't use distortion to agitate—I don't make records for that purpose. I generally want my records to wrap around the listener like a big hug.

That said, agitation is a perfectly legitimate reason to apply copious amounts of irritating distortion.

Edge

There are plenty of reasons to give your record some edge, but mostly it's to introduce some measure of toughness. Distortion that provides edge could possibly cause agitation too, especially if you're liberal with it. You judge that by how irritating you find it. I personally prefer to use distortion for purposes of edge to help toughen up a record that's coming off too sweet.

Sustain

Aggressive distortion can cause a part to sustain, which can be useful on KIKs, snares, toms, bass, and of course, guitars. We love sustain on guitars—it's one of the most desired byproducts of high-gain distortion.

When you push a guitar into excessive distortion, you're technically over-compressing it, which is part of what creates

The Home Producer

Chapter Three Excerpt #3

So, yes, there are technical differences between a maximizer and a brickwall limiter, but they serve the same purpose—to maximize loudness and reduce the dynamic presentation of a mix.

THE STRUCTURAL DYNAMICS OF FLOW

Basic Signal Flow

Before we get into gain staging—the bane of the modern producer's existence (although it needn't be)—we should discuss signal flow. If you understand how audio moves through your system, gain staging becomes far easier to understand.

Signal flow is simply the path audio takes to go from point A to point B. And as with everything in music production—and life itself—what happens now affects what happens down the line.

Once audio is inside your DAW, signal flow becomes more complex, especially since different DAWs handle routing in their own ways. Some even let you customize how the signal flows, which is powerful but potentially confusing. While you'll get a feel for the quirks of your own system over time, it's important to understand how signal flow generally works.

In most DAWs, recorded audio on a channel flows through the inserts first, then to the fader, and finally to the pan knob, before reaching the output. This setup ensures that any processing—such as compression, distortion, or EQ—is applied *before* you adjust the channel's level.

That's important, because if the fader came before the inserts, any volume changes would affect the signal level going into your processors—and that would alter how they behave.

The Many Roles of the Producer For-Hire

Chapter Four Excerpt

feel, what kind of physical reactions they cause. This sort of positive feedback is a great way to reset and can often serve to inspire your talent enough to get you through the tough times.

Track review is also a good idea when you're the one struggling. At the very least, it gives you a basis of comparison and a break from the grind.

Visionary In Chief

Vision, quite simply, is your ability to picture the final product before recording a single note. Vision alone is your most powerful asset as a producer, and more specifically, as the leader of a project. Without vision, you leave everything up to chance. If you don't know how to present an act, if you don't have a clear concept of the project at hand, then you have no business producing it.

Whereas you can produce a record with large gaps in your knowledge base, you cannot produce a record without a vision. Certainly not with any kind of consistency, and probably not well either. As we've already established, an Engineer Producer could have minimal knowledge of music, and a Musician Producer can be weak in the engineering, and both of them can produce an effective record. Those gaps can be compensated for by employing others. Who are you going to hire to fill in for your lack of vision? A producer? *That's you*!

I can promise you this: if you don't like the act, the song, or even the genre, then you will have no vision in regard to the project. You can't possibly. You can't lead if you don't know where you're going, and you can't rightly know where you're going with a production if you don't adore the material. And I mean *adore.*

Clearly, your vision for a project is not so clear that you can predict, with absolute accuracy, the final result. There are far too many variables for that. The free will of your talent alone would

make that impossible. So, vision isn't some form of clairvoyance. Vision has to do with the basic conceptualization of the album, in both sound and approach.

Your vision should take the genre into account. A hard rock band requires a different approach than a pop artist. While the genre will dictate many of the conventions you adhere to, music is far too complex to stay rigidly within the stereotype of a genre. Oh, believe me, there are plenty of producers out there capable of producing only one genre, and who seemingly make the same record over and over again. How boring is that? Your goal is to create something stunning and new, but also recognizable. To accomplish that goal, you need to operate within and outside of convention simultaneously.

Admittedly, there are some genres in which to break from convention in terms of production aesthetic is to be viewed as an almost amateurish and out of touch presentation. Genres can get into ruts so deep that they almost become comical in their adherence to current trends, such that any deviation puts it into a new classification of genre. If that describes your favorite genre, the scope of your vision will be limited to those accepted conventions.

The recording process itself can and will carry great influence over the final product. Thirty days of recording time will yield a considerably different record than one completed in just 10 days. Unfortunately, if you have an act that requires 30 days of polish, and a budget that barely covers 10 days, it becomes impossible to accomplish your vision. In other words, vision dictates budget, and budget dictates vision, and if you can't reconcile the two, you can't make the record. Not properly, anyway.

Now, you don't typically conjure your vision out of whole cloth. All the information is there for you in the form of an act's past records, demos, and live performances. You just have to figure out what it's all telling you. This is true even if an act's

The Business

Chapter Five Excerpt

me for my services than take that kind of risk. If you want to shop the Band—that is, pitch them to labels or managers—you can have an attorney draft a separate agreement for your shopping services, which pays out in the form of points or publishing. However, unless you're well connected, you'll likely need to give up a portion of your points or publishing to incentivize an attorney or manager to shop the act on your behalf.

The good news is, many records can be made fully in your own room, so you can often fund the making of a record with nothing more than your time. That doesn't mean you don't have costs associated with the project, including the attorney's fees and travel costs associated with shopping the act. That alone can get quite expensive.

Preparing the Budget

The cost to make an album can't be left to chance, and you will need to put together an itemized budget of costs. For all intents and purposes, it's a plan, which estimates costs based on your experience making records.

Let's say we're recording a somewhat successful Band and the proposed budget to make the album is $60k. While $60k may seem like an enormous budget to many of you reading this book, it's not an inordinate sum of money for a project that requires studio time and personnel.

The more that you charge as a Producer, the bigger your budget needs to be. Given a $60k budget, it's not unreasonable for me to start out at $3k for my Producers fees. Rather than to limit myself, I'll put in all the costs at a high level, understanding that I will likely blow past our budget in the process. This way, we can discuss strategies for paring the budget down.

Studio Projects

Chapter Six Excerpt

Vocals in the Studio

It's perfectly fine to record a vocal in the studio if we're capturing a performance. Once the process gets more involved than that, it makes more sense to record in a less expensive setting.

When recording in a studio I personally prefer to bring the singer into the control room with me. Not only does it make communication more efficient, but it also allows me to kick everyone else out of the room. If the singer responds to an "audience," as it were, then others on the team are welcome to stay. But the moment they become a distraction, everyone has to go.

Of course, if your singer prefers to perform in the live room, then that's where you should put them. Just be sure you're rolling the moment they're ready to sing. This is not the time to allow your recordist to go into anal mode. Set the preamp and the compressor, make sure you have good level, and roll.

An important part of a producer's job is to keep the process invisible. Performance is about being in the right headspace. When the technology gets in the way, you've failed a fundamental part of your job. Hey, whatever—dust yourself off, and do better next time. Even the most veteran producer will be caught unprepared occasionally.

For the most part, I find it's best if I don't interrupt a performance for notes and suggestions. I'd rather let the singer sing the whole track down and then offer my notes between takes. That said, if I'm not getting anything usable from a take, I don't allow them to struggle. That will only generate frustration. Sometimes people just need a reset.

Encouragement is surely a useful tool to keep your performer's spirits up, but false praise can do more harm than good. It's okay if you have nothing to say beyond, "cool, let's make another one." Believe me, it's better to say nothing than to spit out a bunch of crap that could easily be misinterpreted, all

because you're not comfortable with a little dead air. Both you and your artist are there to work. They seek your expert guidance to help them deliver their best. If you have nothing to add at that moment, it likely means you're not sure of what you think. Wait until you are.

When comping a vocal, tuning is typically my last concern, but it is certainly a consideration in the selection process, especially if I want the tuning invisible. The goal is to assemble a *performance*—one that I can emotionally connect with and keeps my attention. The tuning of your singer can have some role in that connection, although I can promise you, a great performance can withstand some klunkers.

Typically, I send the performer out of the room while I comp the vocal, but it really depends on the singer. Some artists prefer I make the take alone, which is probably the most efficient way to work. Others prefer to be there for the decision-making process. I'm fine with either unless the singer starts working against me, at which point I'll tactfully request some time to make the comp. I may couch it as needing them to stay fresh.

Selecting the best takes for a comp is not something everyone can do well, especially when it's their own performance. Singers can focus on all the wrong things when it comes to their vocal takes. If the singer has good instincts, or is just observing, they're welcome to stay in the room with me. Otherwise, it's best if they get some distance so they can evaluate the compiled *performance* with a fresh ear.

If a vocalist is delivering something great, I don't want to get in the way of that. I'll merely praise them and suggest we record more takes. If they're struggling, I need to figure out why. Is it a monitoring issue? Are they still not fully warmed up? Is their mind elsewhere? Sometimes, they just need to hear playback for themselves, which will help them to make the necessary adjustments. It also allows me the time to formulate my thoughts and offer some direction.

Mixing & Mastering

Chapter Seven Excerpt

Reference Tracks

Many people will suggest you reference other tracks as you mix. This is always about sound. The thinking goes: if you can make your record sound as good as another, then you're good. Some even go so far as to use AI tools like iZotope Ozone to apply another track's EQ curve to their own—which is inadvisable. It's the arrangement and the parts that dictate the overall sonic curve, not how you EQ it.

Referencing tracks for the feeling they evoke is always a legitimate strategy. But referencing sound? That's nothing short of fraught—especially when you're monitoring in a space with serious translation issues. You can absolutely frustrate yourself to no end trying to match tracks in a compromised room. Your references become a moving target.

It's perfectly reasonable to listen to a few tracks at the start of your day, or after a break. That can be an effective way to get your head in the right space. The problem is the direct A/B comparison. For starters, you're probably comparing a fully mastered track to your own work in progress—which is never a fair comparison. You're also likely referencing a song you love, and no matter how hard you try, you can't separate that emotional attachment from your evaluation. The song you love wins every time, which can be demoralizing.

Mixing is a game of confidence, and anything that undermines that confidence will only make your job harder.

Contrary to much of the advice you'll read online, I suggest avoiding direct A/B sonic comparisons and instead focusing wholly on how your mix generates motion and emotional reaction. You really can't match the EQ curve of your track to someone else's—and even if you could, why would you want your production to sound so similar? Your decisions dictate the sound. Trust them.

Mixing & Mastering

Chapter Seven Excerpt #2

For example, if you send a track to an ME where the KIK and snare transients are overbearing, one of the best tools in their kit is a clipper. They'll use it to muscle your balances—in this case, to tame the KIK and snare. And sure, a clipper can help deliver a louder track, but that's only because the balances were out of whack to begin with.

If a simple balance move isn't enough, you can apply more aggressive processing to the individual channels—KIK and snare in this case—to bring them into compliance without introducing distortion outside your sound design choices. You don't need to strap a clipper across your entire mix when you still have internal control.

I'm all for distortion as a sound design choice. That's intentional. It serves a purpose. It's the *added* distortion that's the problem—an artifact of aggressive processing, usually in pursuit of loudness. If you love the result and the distortion doesn't bother you, that's a viable strategy. If not, then you need to seriously weigh the cost-to-benefit ratio of your processing.

These days, I hear about people inserting clippers on their drum bus. I've experimented plenty with clippers—both during mixing and mastering. I've never once felt they improved my mix. But then again, when my balances are off, it's by design.

You can use anything you like. You won't get an argument from me. I'm not handing down rules. "Never use a clipper" is just as ridiculous as "always use a clipper." And yet, you'll see people touting a clipper as a must-have on every production.

In reality, it's a brute-force tool—something you use to muscle balances when you no longer have mix control. If you want distortion on your KIK and snare, do it while you're mixing. Don't wait until the final stage.

Many people insist that you should master the finished mix in a separate session. I bought into that idea for years— mostly because that's how it was always done. Eventually, I found myself constantly bouncing back and forth. I'd apply a mastering

chain, hear a problem, go back to the mix session, adjust a balance, re-bounce, re-import—around and around I'd go. It was an illogically inefficient way to work.

When the technology and the process change, so should the strategy. Once my computer could handle the processing, it made way more sense to apply the mastering chain directly on the mix session itself.

You can work in an archaic way because that's how it was done—or you can strategize based on how you *actually* work. I recommend the latter.

Loudness

There are countless people convinced that loudness offers a value added to their mix, but that is easily proven untrue. If you master the same song twice—once at -7 LUFS and again at -14 LUFS—and then level match them, you can randomly cut between the two and the edits will go by seamlessly as it plays back. I even made a YouTube video proving this called *Let's Talk LUFS*. The -7 LUFS mix had obvious added distortion (which is a degradation, not a value added), and even then, most people wouldn't notice the switches. That's because loudness is not about what happens now. It's about what happens over the course of time. Loudness is all about setting your mix's overall dynamics.

That's why LUFS are measured as integrated loudness. It doesn't matter how loud one section is—what matters is the loudness of the track as a whole. Streaming platforms use integrated LUFS because it gives us a picture of the overall dynamic.

Just as a sheer matter of logic, if we measure loudness as the integrated dynamic range of the entire track (and we do), then a louder track will have less variance in level between sections than a more dynamic one. If you have a 24-bar breakdown section that drops way down in level, that will have a significant influence on the measured LUFS of your track.

The Politics of a Band Entity

Chapter Eight Excerpt

You should avoid voting positions that are not in your favor. Producing can't be done well by committee, so don't put yourself in a position where you only get one vote out of many. Issues will come up, and if most of the band is unhappy, you really have no choice but to find an alternative. If that doesn't work, then you should probably defer the problem.

One of the more compelling reasons to cozy up to the Child of Brain in the first place is to help you navigate the politics of the band. It's not so you can give yourself permission to treat everyone else like second-class citizens. Band members are just as aware of the natural pecking order as you are. Given this, you should go out of your way to give them a sense of importance. If the entire band isn't performing at their best, your production will suffer.

The Art of Presentation

Your best weapon against a political crisis is presentation. How you pitch an idea often matters more than the idea itself. Sometimes, a bold, unexpected suggestion is just the jolt a session needs—especially if you're trying to reassert control or if the band is particularly open and malleable. Other times, it's better to ease the room into a new idea, especially when there's resistance in the air.

The beauty of ideas is that they spark other ideas. When I know exactly what I'm trying to accomplish, I'll often present a concept with full confidence and little explanation—I'm simply lighting the next fuse. But when I'm unsure, and hoping the band can help shape or refine the direction, I'll present that same idea more tentatively, inviting feedback and collaboration.

A large part of your job as producer is sales. Before you ever step into a studio, you're selling yourself, your enthusiasm, your vision, and your value. That sales job doesn't end once the red light goes on. Any time you face resistance, you'll need to sell

your ideas—and your most effective tool for doing so is demonstration.

The entire process of making a record should be one demonstration after another—all of them leading toward your vision. Don't waste time debating abstract concepts that can be tested in real time. If there's disagreement over direction, try it. Build it. Listen to it. Evaluate it. If your way really is better, it'll become obvious. But if someone else has a strong idea, shutting it down without investigation is shortsighted.

You never want to brush off your band's concerns with vague promises. If you've told them they'll always know where they are in the record, you can't turn around and say, "It'll all work out later." If there's a problem, acknowledge it—even if you're not ready to solve it yet. Deferment is fine; dismissal is not. And whatever you do, don't tell your band you'll "fix it in the mix" (even if it's true). If you can fix it in the mix, chances are, you can fix it right now.

Festering concerns kill morale. Communication keeps it alive. If you're struggling with a track, let the band in on it. They're not just your clients—they're your partners in this. And they might just have the solution you need.

Deferring Decisions

As we've already established, it's generally best to avoid deferring decisions when producing a record. Politically, however, there are times when it's necessary to table a discussion. Whether you're dealing with a split band too entrenched in their positions to negotiate, or an emotionally charged Child of Brain, deferment can be a useful tool. There's no point in allowing a debate between two sides incapable of hearing each other. Use time to your advantage.

The passage of time allows emotions to settle, which makes compromise more likely. You can also use success as emotional leverage to reopen dialogue and solve difficult problems. Success

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